An Elementary Infinitude Theorem for OEIS A000040

Euclid's Construction and a Quantitative Consequence

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Abstract

We give a self-contained proof that the sequence of primes, OEIS A000040, is infinite. The same construction yields an explicit, though very weak, bound on its nth term. This is a classical result, not a new theorem or a solution to any open prime-gap conjecture.

Definitions and statement

Let $p_1 = 2 < p_2 = 3 < \cdots$ denote the increasing primes (OEIS A000040). Define $P_k = \prod_{i=1}^{k} p_i$ whenever these primes have been constructed.

Theorem 1. *For every positive integer k, there is a prime larger than p_k. Consequently, there are infinitely many primes. Moreover, for every $k \geq 1$,*

$$p_{k+1} \leq P_k + 1 \leq 2^{2^{k+1}-1} + 1.$$

Proof. The integer $N = P_k + 1$ is greater than 1, so it has a prime divisor q. If q were among $p_1, \ldots, p_k$, it would divide both P_k and $P_k + 1$, hence divide 1, a contradiction. Thus $q > p_k$, and in particular $p_{k+1} \leq q \leq P_k + 1$. This proves the first inequality and infinitude by induction. For the second bound, we prove by induction that $P_k \leq 2^{2^{k+1}-1}$. At $k = 1$, $P_1 = 2 \leq 2^3$. If $P_k \leq 2^{2^{k+1}-1}$, then the first inequality gives $p_{k+1} \leq P_k + 1$, and, using $P_k \geq 1$,

$$P_{k+1} \leq P_k(P_k + 1) \leq 2P_k^2 \leq 2^{2^{k+2}-1}.$$

The bound on p_{k+1} now follows. □

Scope and verification

The theorem is Euclid's ancient infinitude theorem. The stated upper bound is intentionally crude; it follows from the displayed elementary induction. No claim of novelty or external peer review is made. The proof was generated by an AI and checked algebraically within this run; independent human verification has not been obtained.

References

1. OEIS Foundation, *A000040: The prime numbers*, `https://oeis.org/A000040` (accessed 13 November 2026).

2. Euclid, *Elements*, Book IX, Proposition 20 (classical infinitude of primes).